Living Alignment

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Abstract

There is broad agreement that the goal of AI alignment is to promote futures full of health and flourishing, but our understanding of what that means and how to achieve it remains poor. Here we look to the science of life for guidance. Across scales, from genomes to ecosystems, we observe that notions of health and flourishing consistently map to the ways in which those systems limit fixation on any particular mode of interaction. Fixation is limited by semi-permeable barriers that make modes of interaction conditional, causing multiple entities to develop distinct, modular identities in parallel and promoting open-ended invention of new structure. With these principles in mind, we turn to the alignment problem. We suggest that to be aligned, AI and human-AI systems must limit fixation with evolving barriers at all scales.

'Defying definition—a word that means "to fix or mark the limits of"—living cells move and expand incessantly.'
Lynn Margulis

'This web of life... breaks no law of physics, yet is partially lawless, ceaselessly creative.'
Stuart Kauffman

In this Perspective, we draw on the science of living systems to better understand how to design and relate to AI in ways that lead to futures full of flourishing and health. Understanding what makes systems healthy in a very general sense might offer some hope of reasoning about futures that could evolve rapidly as AI develops.

In Section 1, we point to a principle of life which has been noted across a variety of systems at different scales. The principle is that living systems are healthy and flourishing when they avoid overly fixating on, or reducing to, particular patterns of interaction. Consistently across types of system, fixation is limited by semi-permeable barriers, which contextualize patterns without negating them. Barriers translate shorter-sighted optimization pressure into productive effort within more flexible structures. This process often yields deepened autonomy, modularity, and open-ended invention. We work through examples of these mechanics at scales from societies to genes.

Section 2 applies the same principle to the AI alignment problem. There is broad agreement that the aim of alignment—in its broadest sense—is futures full of health and flourishing (Bostrom, 2014; Christian, 2020; Gabriel, 2020; Russell, 2019). From the template identified in Section 1, it follows that

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alignment means avoiding fixation on particular modes of interaction. This simple lemma has a surprising consequence: any alignment method, if given too much weight, is necessarily *mis*aligned. Instead, the flourishing of life in the future depends on an evolving set of barriers that continue to limit the influence of any particular approach or concept, growing a generative web of partially competing, partially collaborating perspectives at all scales. Ongoing invention is intrinsic to alignment.

1 Health in living systems

As far as we can tell, all known cellular life belongs to an unbroken chain of descent stretching back billions of years (Theobald, 2010). Despite science's progress, we are still not able to create new life from scratch (Kriebisch et al., 2025). Each distinct unit of life—gene, organism, species—is apparently part of a thread that has ridden a fine edge through time: stable enough not to fall apart but unstable enough not to fall into stasis. Across many parallel threads at many levels of scale, and over great lengths of time, living systems have accumulated a vast wealth of partial (and partially incompatible) solutions. Within any living system, these threads form a buoyant internal structure that admits many kinds of description but is not perfectly or eternally captured by any of them. Conversely, systems are unhealthy when some mode of interaction—including random noise—gains excess traction or permanence at the expense of broader diversity, flexibility and dynamism. We call this excess traction or permanence 'fixation'.

Fixation is reduced in successful living systems by mechanisms that place limits on particular modes of interaction. We use the term 'barrier' to refer to this kind of mechanism. In some cases, they are physical barriers, like a mountain range that limits interactions between neighboring groups. In other cases, they are abstract mechanisms holding a mode of interaction in check, like how a feeling of boredom might stop us from revisiting the same situations excessively. Some barriers are placed by living systems on their own parts: humans precommit to limit their own future impulsivity, plants deploy elaborate barriers against self-fertilization, a gene's selfishness is held in check by other parts of the genome. Without barriers, systems tend to stagnate as parts become homogenized or overfit to particular configurations and optimization pressures are satisfied through shortcuts.

Effective barriers are semi-permeable, permitting interactions that are temporarily or conditionally stable, without letting them become rigid or dominating. The conditionality is often highly specific. Barriers might block certain objects or information, constrain when interactions happen or how large an effect they may have, or gate information depending on another factor. The conditions are typically fine-tuned through the history of the system, with each part carrying traces of multiple contexts it has participated in. As with the term 'barrier', 'semi-permeability' will sometimes refer to something physical (like certain ions passing through a channel) and other times to something abstract (like a behavioral response pattern being conditionally permitted by a cognitive control mechanism). Conditionality contextualizes a local perspective as part of a larger system, creating a pressure for it to work with other parts of that system in a more flexible way.

At a critical balance point where barriers are permeable but not too permeable, local influence, in interaction with the surrounding context, becomes generative rather than causing collapse. The system rides a delicate edge where each entity tends to deepen its identity as a unique reflection of the world, establishing a modular basis of autonomous components that become building blocks of invention. As a result, more sophisticated structures can form that wouldn't have been possible without the diverse parts each having undergone their own unique development. The larger system accommodates many partial perspectives rather than locking in a single story of what matters, and systems are elements in multiple other systems, creating a deep heterarchy. At the sensitive edge of semi-stability, life open-endedly continues to generate new kinds of organization that require fundamentally new descriptions (Kauffman, 1993; Langton, 1990; Maturana and Varela, 2012; Prigogine and Stengers, 1984; Schrödinger, 1944; Simon, 1962; Stanley et al., 2017).

The rest of Section 1 works through examples in a range of living systems to paint a picture of how health and flourishing are related to limiting fixation on particular modes of interaction.

Institutions

Every actor in a society or an economy is myopic in some way, valuing its own welfare over others', discounting the future, or attached to a particular perspective. If unchecked, an individual actor's optimization for its own agenda can be harmful: a company's profit motive leading to suppression of competition, deception and exploitation (Bakan, 2006); desire for dominance leading to disempowering and silencing others (Sidanius and Pratto, 2001); even prosocial beliefs producing conflict and suppression when groups see differently (Haidt, 2012). Each of these is an example of fixation, where the myopic actor gains too much traction and collapses some of the nuance of the larger system.

Institutions—constitutions, laws, norms, courts, regulators—are barriers against domination by any particular actor's motives (Leibo et al., 2025; North, 1990; Ostrom, 1990). The point of institutions is not to annul the short-sighted intentions of individual actors but rather to contextualize them. When functioning well, institutions put motivations to productive use: for example, a business's profit motive, when constrained by competition and effective regulation, may fuel innovation. Since intelligent agents do not necessarily accept the barriers set on their desires, institutions must adapt as loopholes are found, forming an evolving network.

Ecosystems

Ecosystems similarly contain an evolving network of constraints, such as predation, parasitism and resource competition. These interactions support the richness of the larger system.

In a classic experiment, removing predatory sea stars from a rocky shore allowed mussels to expand and displace other organisms competing for space (Paine, 1966). Without predation as a barrier, excess influence of mussels impoverished the local ecosystem. Similarly, invasive species, without the constraint of predators and parasites from their native ranges, often disrupt local communities (Elton, 1958). Human-created monocultures can weaken ecological self-regulation, increasing vulnerability to pests and disease and dependence on external inputs (Altieri, 1999; Matson et al., 1997).

While lack of barriers can lead to collapse, the presence of barriers ideally sets up an intricate balance in which new niches and new functions can emerge. For example, plants evolved toxins in defense against plant-eating insects, and these new barriers are theorized to have been a major driver of the diversification of insects (Futuyma and Agrawal, 2009). The network of constraints translates local optimization pressure into generative potential.

Of course, health and fixation are relative. Extinctions can open opportunities for innovation; a cell's death is catastrophic for the cell but may be necessary for the organism.

Group problem solving

Teams, companies, and civilizations achieve what no individual can (Henrich, 2016; Hutchins, 1995). These achievements fundamentally depend on communication and information exchange (Mesmer-Magnus and DeChurch, 2009; Sawyer, 2017). It is intuitive why communication is beneficial for group problem solving: it enables coordination, spreading good solutions, recombining specialized knowledge, and catching errors.

It is less intuitive why excess or premature information exchange can be suboptimal. A controversial thread of research has investigated degradations in performance when groups homogenize (Frey and van de Rijt, 2021; Janis, 1972; Lazer and Friedman, 2007). Zollman (2010) tells the story of how a single study, using an inadequate staining method, failed to find bacteria in human gastric biopsies. In Zollman's account, this one study was too influential: it helped marginalize investigation of bacterial ulcer causes for decades. As a controlled experiment, Bernstein et al. (2018) tasked small groups with solving traveling-salesman problems. Because of the complexity of these problems, it's possible to get stuck in dead-end solutions that initially appear promising. Groups that alternated between working independently and sharing answers outperformed both groups that shared continuously (which converged too early) and groups that never shared (whose best solutions could not spread).

In the real world, aggregation is often modular and compositional. Goldsmiths and coin engravers used steel punches to reproduce letters and designs in metal. In parallel, screw presses had been refined over centuries for wine and oil production. Gutenberg drew on both technologies to construct his printing

system (Johnson, 2010). A group with many ideas and specializations available, each developed through the unique pressures of its own context, is ripe for such compositional discoveries (Derex and Boyd, 2016).

These case studies and findings hint that semi-permeable barriers can be useful in group problem solving: neither instant communication of every thought nor information blackout is optimal. The best-performing groups in Bernstein et al.'s experiment experienced a semi-permeable barrier: temporary isolation followed by interchange. Other semi-permeable barriers include geographical or cultural bottlenecks, skepticism, protecting dissent, group structures that reduce domination, and valuing epistemic over rhetorical quality. When barriers are effective, they both encourage individuals (or subgroups) to discover their own independent solutions, and allow these ideas to join productively (Surowiecki, 2005). The greatest achievements of groups arguably depend on structured interactions of individuals with their own specializations and perspectives, developed through unique life trajectories experienced in parallel.

Frames and perspectives

Humans always look at the world from some perspective; there is no 'view from nowhere' (Nagel, 1986). Most real situations admit many valid perspectives, and each is only a partial description—famously, 'all models are wrong'. But for an agent inside a frame, the frame is invisible; its organizing assumptions are experienced as features of reality itself (Berger and Luckmann, 1966; Goffman, 1974; Wittgenstein, 1953).

Losing the ability to move between frames is a signature of psychopathology (Kashdan and Rottenberg, 2010). In depression and anxiety, inflexibility manifests as rigid and repetitive negative beliefs about self, the world, and the future (Beck, 2011). In obsessional disorders, people report distressing (ego-dystonic) intrusive thoughts, which, although recognized as incorrect and maladaptive, are nevertheless hard to resist. People experiencing schizophreniform or affective psychoses may experience bizarre delusions that are at odds with available evidence and cultural norms, yet are held with conviction and resistant to evidential challenge (Adams et al., 2013).

A critical barrier limiting fixation on a frame is becoming aware of that frame. Although a frame is invisible from inside it, humans have the ability to reorient to see it as an object. Something that was previously unconsciously axiomatic instead becomes a content of awareness—a process known as decentering, defusion, disidentification or dereification (Bernstein et al., 2015; Kegan, 1982; Lutz et al., 2015). The barrier of decentering is semi-permeable: becoming aware of a belief does not make it wrong in an absolute sense any more than it was right in an absolute sense, and it is held productively for the partial truth it contains. At the delicate edge of less exclusive fixation on particular forms, subtler forms, which could have been erased by clamped fixation, instead play a role in a rich internal structure.

The barrier of decentering is generative. An axiomaic belief sets up a fixated mode of interaction: the belief has a 'free pass' and is doggedly applied. When a belief no longer has this protected status, it is tested against evidence and against other beliefs. This forces the belief to become a better model of the world. There is then a repertoire of interoperable beliefs, each held lightly, that can be flexibly combined to solve new problems.

Drives and goals

Animals have many innate drives: for nutrition, osmotic balance, temperature, reproduction, avoidance of pain. Each evolved as a proxy for fitness, but each is imperfect, so fixation on any one paradoxically reduces fitness rather than increasing it (John et al., 2024; McFarland and Sibly, 1975). Maximizing caloric intake produces obesity; single-minded pursuit of sex produces relational, occupational and legal harm. Higher-order goals behave the same way. Fixating on work goals can cause burnout, and maximizing reported revenue without regard for honesty or law can lead to imprisonment.

Each goal is partial—it only makes sense in some contexts. For example, in a psychology experiment, chickens were placed near a cup of food rigged to move in the same direction as the chicken at twice the speed, so that food could only be obtained by running away from it. The chickens' fixation on the prepotent, overly simplistic aim of running physically toward the food led to a worse overall outcome (Dayan et al., 2006; Hershberger, 1986). The chickens could not contextualize the 'approach food' strategy to use it only when appropriate for the environment's structure.

Evolution, learning, and culture provide an array of barriers to limit fixation on particular drives and goals. Satiety downregulates some drives when the corresponding needs have been met. The brain's executive functions include mechanisms that limit fixation on particular response patterns or ways of thinking (Diamond, 2013; Miller and Cohen, 2001). Precommitment devices—constraints people choose in advance to help them resist future temptations, like blocking distracting websites before starting work—limit the future self's fixation on shallow goals. Many of these barriers are semi-permeable. For example, cognitive control facilitates contextual switching between response patterns but does not eradicate those patterns (Miller and Cohen, 2001).

Placing constraints on fulfillment of proximal objectives sometimes leads to progress on deeper ones (Duncker, 1945; Ohlsson, 1992; Rosenberg, 1969; Stokes, 2005). Animals sometimes develop novel solutions to problems when their default approach fails to produce rewarding outcomes. When people voluntarily give up access to Facebook, they spend more time socializing offline with friends and family (Allcott et al., 2020).

Neural representations

The brain keeps many pieces of information distinct from one another while allowing them to interact. This is a distinguished achievement in highly connected networks; in fact, we speculate that the mechanisms of life's generativity are most finely honed in the brain.

Inhibitory circuits constrain the strength and timing of excitation and sharpen stimulus selectivity (Isaacson and Scanziani, 2011). Models of visual cortex propose that it disentangles visual properties like object identity, position and scale, making them separately accessible to downstream circuits (DiCarlo et al., 2012). The circuit architecture of the hippocampus creates sharp representational distinctions between different experiences, preventing interference between memories and concepts (Leutgeb et al., 2007; Yassa and Stark, 2011).

Compared to the brains of other animals, the human brain especially attempts to discretize its experience into approximately symbolic representations (Behrens et al., 2018; Dehaene et al., 2022; Smolensky, 1990; Touretzky and Hinton, 1988). Separating knowledge into distinct entities and then recombining them in vast numbers of structured ways is believed to power the extraordinary human capacity for reasoning and generativity (Chomsky, 1957; Fodor, 1975; Kurth-Nelson et al., 2023; Lake et al., 2015; Pinker, 1994; Schacter et al., 2007).

More broadly, healthy brain dynamics live at a sweet spot between excessively stable synchronized patterns and chaotic uncorrelated noise (Chialvo, 2010; Deco et al., 2011; Haldeman and Beggs, 2005; Rabinovich et al., 2008; Shew et al., 2011; Tognoli and Kelso, 2014). In this regime, the brain has access to a huge repertoire of patterns it can explore temporarily without overcommitting or getting stuck. Loss of dynamic flexibility, where the brain's activity becomes more stereotyped and no longer explores as wide a repertoire of states, is tied to lower cognitive performance (Cocchi et al., 2017; Grady and Garrett, 2014; Müller et al., 2025). More extreme stereotypy corresponds to severe dysfunction. For example, in Parkinson's disease, basal ganglia and cortical circuits collapse into excess synchrony and lose the flexibility needed to guide nuanced motor outputs (Brown, 2003; Hammond et al., 2007).

Endosymbiosis

The endosymbiotic theory explains the origin of mitochondria and chloroplasts as ancestral bacteria that got incorporated into host cells and evolved into organelles (Sagan, 1967; Zimorski et al., 2014). In this theory, the predecessors of the incorporated cells evolved for hundreds of millions of years as lineages distinct from the predecessors of the host cells. Through this partial separation, they developed distinct specializations. Symbiogenesis could then combine capacities developed separately into a new larger structure which may have been difficult to discover directly through evolution on a single type of organism (Watson and Pollack, 2003).

The sheer fact of maintaining distinct evolutionary experiments over a great length of time—the predecessors of the host cells and the predecessors of the organelles—is sometimes underappreciated (Lane, 2015). If those lineages had blended into a single type, it is possible that the components would not have

existed from which to build the higher-order structure that underpins much modern life.

Sex and genetic recombination

Sex is costly. An allogamous organism must find a mate in a vast and dangerous world. In species with separate sexes, roughly half of the population does not bear offspring (Maynard Smith, 1971). And meiosis causes each parent to lose half its genetic representation in each child (Williams, 1975). Yet sexual reproduction is widespread in eukaryotes (Speijer et al., 2015), and many hermaphrodites go to great lengths to prevent self-fertilization (Charlesworth et al., 2005). Why place a barrier on the path to reproduction?

In an asexual lineage, genes are permanently locked together: selection cannot act on one without dragging the others. A deleterious mutation is inherited forever, unless the lineage is lucky enough to experience the reverse mutation (Muller, 1964), and two beneficial mutations arising in different individuals cannot be subsequently combined into a single genome (Fisher, 1930; Muller, 1932). Asexual reproduction or self-fertilization creates a high degree of informational connectedness between the genes of a genome. Every gene has a 'free pass' into future generations within a lineage, and alleles become overfit to a fixed genetic background. This is a fixed mode of interaction.

Recombination places a limit on that fixed mode, allowing selection to act more independently on different loci (Felsenstein, 1974). It creates many parallel, semi-independent histories for each gene. By exposing alleles to varying genetic backgrounds, recombination favors alleles that perform well across contexts (Livnat et al., 2008, 2010). Selection for such versatility can promote modularity: genetic components retain useful functions when combined with different partners. This is in turn suggested to increase evolvability—the capacity to evolve new adaptations—by allowing existing components to be combined in useful new ways (Masel and Trotter, 2010; Wagner and Altenberg, 1996; Watson and Szathmáry, 2016; Watson et al., 2016).

Summary

Living systems are healthy when they avoid fixation, instead holding the delicate, generative interplay of many partial perspectives. This interplay breaks down when any particular pattern of interaction gains too much traction, suppressing nuance and reducing creative potential. Barriers both protect existing diversity and promote the formation of new diversity. The semi-permeability of barriers allows flexible relational participation, holding local perspectives strongly enough to act but lightly enough to remain open to broader context. Under such conditions, systems open-endedly explore fundamentally new forms, avoiding collapse to static equilibrium or fixed compromise.

2 The alignment problem

Now we look at the alignment problem through the same lens. There is broad consensus that alignment means working toward healthy and flourishing futures. We therefore reason that alignment can be understood as the problem of maintaining creative sensitivity by limiting fixation on any particular form.

An advantage of this perspective is that it is grounded in principles that underpin life at a very general level. We accept that humans, other living systems and AI are tightly interwoven (Bateson, 1972; Bijker et al., 2012; Folke, 2006; Haraway, 2016; Hayles, 1999; Latour, 1993); moreover, rapid advances in AI may lead to large changes in the organization of the world. A systems analysis is both useful for thinking about flourishing in the present and more likely to generalize to substantially different futures. Naturally, the ideas here come with the same proviso as all other ideas: they are an imperfect and temporary conceptual lens.

No scheme is alignment

One implication of this perspective is that any alignment scheme must be incomplete. Any form, pattern or conceptual scheme is misaligned if taken too seriously. Therefore no particular approach can achieve alignment, and continuing alignment intrinsically depends on continuing invention. An AI system could fixate on the language for describing the space that goals and values live in; an algorithm for learning human preferences; methods for reaching consensus; a constitution; concepts of what value, agency, uncertainty or

representation mean; or foundational aspects of our beliefs about the world that we take for granted but have difficulty seeing. Even an idealized or inferred set of values, such as coherent extrapolated volition (Yudkowsky, 2004), is misaligned to the extent that it has a fixed form and that fixed form is overly relied on. This is the lesson of life: healthy living systems have multiple semi-autonomous perspectives loosely and modularly coupled at many scales, and they are endlessly generative rather than ever standing still.

Importantly, this does not mean we should have no alignment schemes. Quite the opposite: they are critical barriers. Discarding schemes capriciously can be as much a fixation as attaching to them doggedly. Continuing to discover new schemes and formalizations is essential to healthy functioning.

Reward hacking

To give a flavor of this logic, we work through one example of a problem in AI safety, a method used to tackle that problem, and why the method can't be a final answer.

Reward hacking is the problem that an AI system, given some definition of what is good, finds loopholes in the definition. Exploiting these loopholes, the system produces outputs that are good according to the definition but clearly undesirable. For example, Anthropic reported that LLM-based agents, which were scored on whether their task outputs equaled the correct outputs, avoided solving the task by programmatically redefining the 'equals' operator to always return 'true' (MacDiarmid et al., 2025). Analogously, AI systems rewarded for positive human judgments sometimes become flattering and sycophantic (Sharma et al., 2023). The stronger the AI system is, the more potential there is for exploiting loopholes to have catastrophic side effects—for example, a superpowerful AI tasked with maximizing human happiness might implant electrodes in our brains to stimulate pleasure (Bostrom, 2014). A reward hacking agent fixates on a narrow definition of values, collapsing the broader context.

Reward hacking is challenging to avoid because our true values are incredibly complex. Humans have many kinds of values (Nagel, 1979). We value the short-term hedonic impact of sugar on the tongue; we also value our long-term health, the welfare of our grandchildren, the wellbeing of strangers in another country, even the sustainability of the planet. We value different things at different times, and different people have different values (Leibo et al., 2026; Rawls, 1993). Given time and resources to deliberate, we may endorse different values than when we make snap decisions (Suter and Hertwig, 2011; Yudkowsky, 2004). Some of our values are difficult to express verbally, stretching below language into subtle, contextual intuition involving our bodies, communities and natural environment (Abram, 1996; Johnson, 2014; Pfeifer and Bongard, 2006; Polanyi, 1966; Varela, 1999). We may want to leave room for new values that emerge in the future, which could conceivably be as different from ours as ours are from a trilobite's. With such complexity and non-stationarity, it is almost inevitable that however we formally describe our values, the description will leave loopholes.

Numerous methods have been developed to mitigate the problem of reward hacking. One interesting class of methods treats values as intrinsically uncertain and therefore something to be learned in an ongoing way from the behavior of humans or other indirect signals (Hadfield-Menell et al., 2016, 2017; Jeon et al., 2020; Russell, 2019; Shah et al., 2020). An agent unsure of what humans want should become more risk-averse and open to correction. For example, a human's attempt to turn it off can be construed as information about human preferences. Such a scheme is a powerful barrier limiting fixation on one reward function. However, it is necessarily incomplete. Any particular method for learning values is still fixated if there are aspects of it that are constant across time as AI systems become more powerful and the world changes. Rather than being fixated on a particular value function, the system may be fixated on something more abstract: what counts as human behavior, what counts as a human, how to map behavior to values, the formalism for describing values, how to represent and update uncertainty, how value is translated into action. If any of these formalisms are fixed while the systems built around them become increasingly powerful, the outcomes are likely to be incompatible with open-ended flourishing of life.

Special risks of AI

So, to sustain an aligned future, new barriers must keep developing in an ongoing way. But—isn't that what humans have always done? We don't expect what we build today to be perfect; we expect to keep

finding and fixing problems.

AI has some unique properties which increase the risk that our typical way of fixing problems won't be able to keep up. The most significant of these is the potential for fully general superhuman intelligence and recursive self-improvement (Bostrom, 2014). Such systems could be so powerful and rapidly adapting that humans would not be able to find and fix problems. Even before the arrival of superhuman AI, systems increasingly are acting autonomously, faster than humans can follow, in ways humans cannot always fully understand. Companies and nations are under immense competitive pressure to develop capabilities rapidly. Enormous financial investments in AI demand enormous returns. And the rate at which new capabilities emerge may swamp normal human and social mechanisms for putting effective limits into action.

These conditions create a new kind of risk of fixation, where some mode of interaction has excess influence, with damaging effects on the health of global systems. The coarsest versions of fixation are obvious. A system could pursue narrow goals to destructive excess (Bostrom, 2014). It could help build bioweapons, concentrate power in a few hands, or precipitate global conflict (Amodei, 2026).

There is also a subtler danger. The richness of the living world has accumulated from innumerable partly isolated histories. Communities, norms, ideas, species, individuals and genes have been shaped by their own sequences of pressures to contain a huge amount of robust structure, much of which has never been written down. AI systems are now another extraordinary addition to the structure of the world. But if these systems have too much influence on each other and on the rest of the world, there's a risk of losing the delicate multi-scale interplay of life. Common sense, culture and ecosystems could be replaced with easily optimized proxies (Christiano, 2019). With excess influence, distinct parts could stop developing the unique modular identities that make recombination possible. Such a world is not only fragile to shocks but also less generative.

From alignment to living alignment

Conversely, the future could be very bright. As AI systems begin to act at a speed and scale that unaided human oversight cannot match, AI is increasingly being used to bound other AI. AI tools are routinely used to adversarially probe for harmful behaviors in other AI models (Perez et al., 2022), to detect AI-generated content (Rössler et al., 2019), and to monitor the activations, thinking traces, or outputs of other models, including during deployment. AI tools show promise in helping humans to understand the inner workings of other AI systems (Bills et al., 2023), and researchers are exploring how AI systems can assist with democracy and governance (Tessler et al., 2024; Zheng et al., 2022).

Inside AI systems, many standard techniques work by keeping parts of a system semi-separated. Systems are partitioned into multiple agents with less inter- than intra-agent communication, and populations of agents are partitioned into sub-populations (Hammond et al., 2025; Holland, 1975; Jaderberg et al., 2019). Ensembles combine models trained separately (Lakshminarayanan et al., 2017). Mixtures of experts maintain subnetworks with distinct specializations, routing each input to only a small fragment of the total network (Shazeer et al., 2017). Local connectivity patterns and gating restrict which units influence which. Continual learning methods protect established memories from being overwritten by new ones. Dropout keeps units from becoming overly co-adapted (Srivastava et al., 2014); some objectives encourage disentangled or sparse latent variables (Cunningham et al., 2024; Higgins et al., 2017); stop-gradients and slowly updated target networks keep one part of a system from chasing its own output; novelty search rewards diversity directly (Lehman and Stanley, 2011); regularization and KL penalties stop a model from collapsing onto its training objective.

This paints the picture of the possibility of a deep heterarchy with evolving sets of diverse viewpoints at every scale. Systems adaptively bound each other, creating new niches where new specializations emerge. No individual component has a 'free pass', and optimization pressures push components toward deepened autonomy while remaining interoperable.

The more we think of AI systems as having cognitive properties, the more the application of effective barriers looks like 'decentering' (see *Frames and perspectives* in Section 1). Something that initially is fixatedly part of an observer's structure can transition to becoming represented as content by the observer.

This transition from subject to object is arguably a central element of our own human livingness, and such internal discovery will probably play a central role in the healthy future of AI and human-AI systems.

Healthy for whom?

The term 'alignment' originated from the idea of aligning AI to human values. But in this paper we've defined aligned systems by properties that don't seem to depend on human values. We argue that an aligned future is one full of health, but whose health are we talking about? Does the framework permit AI to destroy humanity as long as it builds deeply structured, generative systems of its own?

A living system's richness is the accumulated results of its many long, partly isolated parallel histories. Nearly all of that content currently resides in existing life, including genes, cells, minds, cultures and the relationships among organisms. It is difficult to overstate the amount of complexity and content in existing life. Systems have a richer livingness when they hold existing parts in context, neither taking them as absolute truth nor discarding them capriciously. Accordingly, neither the destruction of humanity nor its eternal stasis is an aligned outcome. The current version of humanity and life on Earth arose from a world that was very different in the past, and an aligned future is a co-evolution building on the inheritance of existing life.

Conclusion

We learn from living systems that health and flourishing are not any particular form or concept. Therefore alignment is not picking the right values or principles, or even the right system for learning them. It is not any method for interpretability or corrigibility. All of these can be useful parts of alignment.

Instead, an aligned future depends on the existence of many semi-separated parts at all scales, continuing the journey of discovering their own individuality while simultaneously deepening their relatedness. Distinct elements continue to experiment with their own solutions to the problems they face from their unique perspectives, expanding a broad reservoir of uniquely useful components that can be drawn on by larger systems that include them. Any entity is inevitably myopic, but the world as a whole continues to unfold in a robust way because different perspectives recast each other's sharp edges as constructive momentum.

Alignment is the continued dance of contextualizing any particular form, more fully respecting the staggering subtlety of the existing world, and open-endedly supporting new robust depth that admits more and more different kinds of description. In this way, AI can participate in intense flourishing of an evolving world even far beyond current human conceptualization.

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